

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #36 — Retail, E-Commerce & Finance
Restaurant Table Booking & Pre-Ordering App

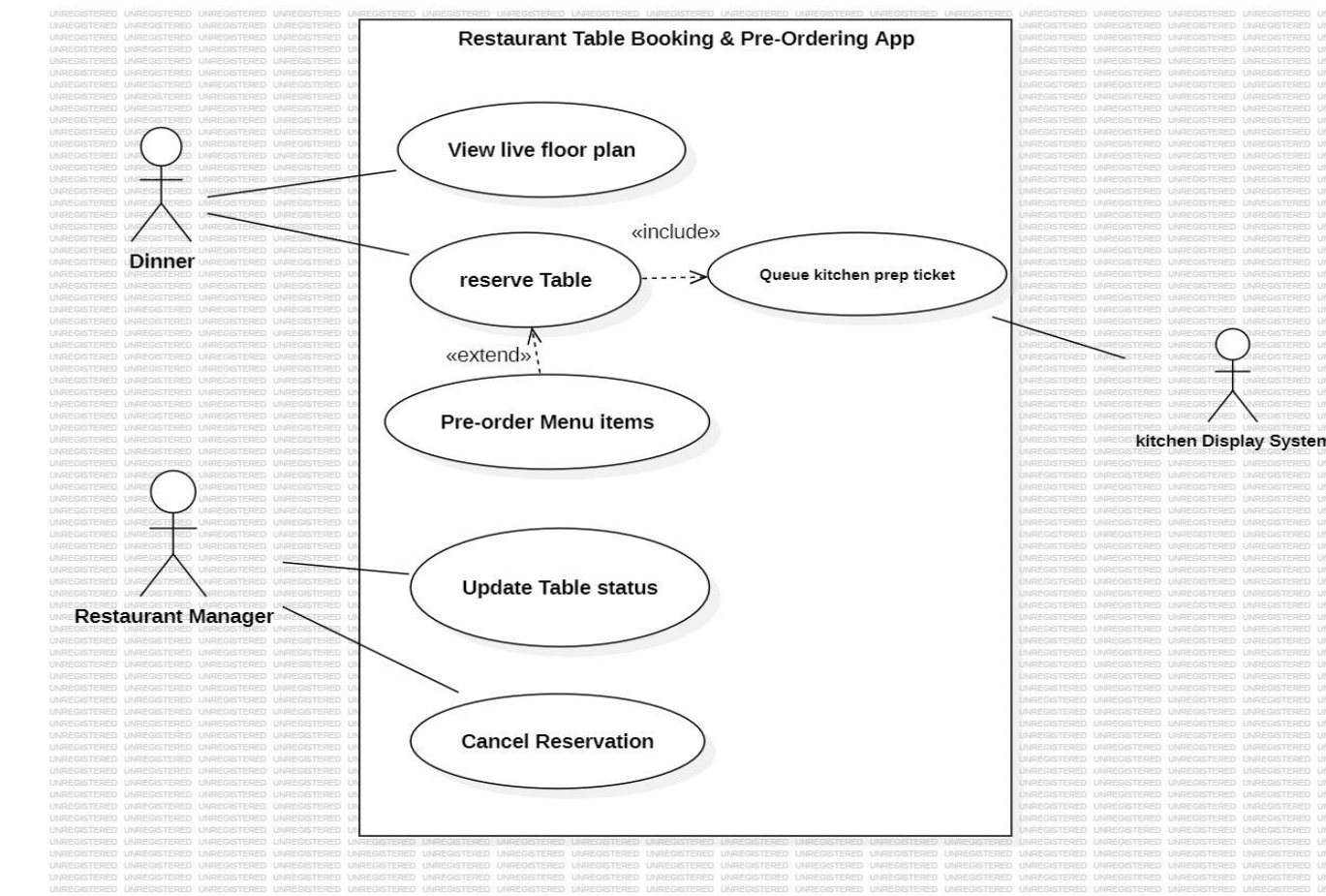
1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	Pass: Table reserved and kitchen pre-order ticket queued. Fail: Overbooking of table during overlapping dining window.	Given
FR-002	Functional	The system shall display a live floor plan to diners showing each table's current status (Open, Reserved, Occupied) before they choose a table.	High	When a diner opens the booking screen, every table renders with a status colour/label matching the backend state, refreshed within 2 seconds of any change.	Diners need accurate real-time availability to avoid selecting a table that is already taken.
FR-003	Functional	The system shall allow a diner to pre-order menu items and link them to a specific table reservation at the time of booking.	Medium	When a diner adds menu items during booking and confirms, the reservation record stores the itemized pre-order, and a corresponding kitchen prep ticket is generated.	Pre-ordering lets the kitchen prepare food in advance so it is ready close to the diner's arrival time.
FR-004	Functional	The system shall allow a Restaurant Manager to update a table's status (Reserved, Seated, Billed) from any floor terminal.	High	When a manager changes a table's status on one terminal, the new status is reflected on all other floor terminals within 500 ms.	Front-of-house staff must see consistent, up-to-date table state to seat and serve guests efficiently.
FR-005	Functional	The system shall allow a Restaurant Manager or the diner to cancel a reservation before the arrival window, automatically releasing the table back to the available pool.	Medium	When a reservation is cancelled more than 15 minutes before the booked time, the table's status reverts to Open and it becomes selectable by other diners.	Timely cancellations prevent tables from being held unnecessarily and reduce lost revenue from no-shows.
NFR-001	Nonfunctional	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given

NFR-002	Nonfunctional	The system shall locally queue and cache kitchen pre-order tickets for up to 10 minutes if the connection to the Kitchen Display System is lost, then automatically resend all queued tickets once the connection is restored, with zero ticket loss.	Medium	When the Kitchen Display System is disconnected and reconnected within 10 minutes, all tickets generated during the outage appear on the kitchen display in original order with no duplicates or omissions.	Kitchen operations cannot afford to silently lose an order during a brief network interruption.
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UML Use-Case Diagram

Actors: **Diner**, **Restaurant Manager**, and **Kitchen Display System** (external system). The diagram below models all primary use cases together with one «include» relationship (Reserve Table always includes Queue Kitchen Prep Ticket, since every confirmed booking must notify the kitchen) and one «extend» relationship (Pre-Order Menu Items optionally extends Reserve Table, only when the diner chooses to order food in advance).



2. Use-Case Flow Specification

Use Case: UC-02 – Reserve Table

Actors	Diner (primary), Kitchen Display System (supporting)
Preconditions	1. The diner has opened the app and selected a valid dining date and time window. 2. The live floor plan shows at least one table with status "Open" for that window.
Postconditions	1. The selected table's status changes from Open to Reserved and is locked against double-booking for the chosen window. 2. A reservation record is created with a unique booking ID. 3. A kitchen prep ticket is queued on the Kitchen Display System, including any linked pre-order items.

Main Success Scenario

- 1 The Diner opens the booking screen and selects a dining date and time. The system displays the live floor plan (UC-01) showing table status for that window.
- 2 The Diner selects an Open table on the floor plan.
- 3 The system displays a booking form and asks whether the diner would like to pre-order menu items. If the diner chooses to, the system invokes UC-03 (Pre-Order Menu Items). «*extend*» 4 The Diner confirms the reservation.
- 5 The system validates that the table is still Open for the requested window (to prevent a race condition with another diner).
- 6 The system marks the table as Reserved and creates a reservation record with a unique booking ID.
- 7 The system invokes UC-04 (Queue Kitchen Prep Ticket) to send the booking, and any pre-ordered items, to the Kitchen Display System. «*include*»
- 8 The system displays a confirmation screen to the diner with the booking ID, table number, and estimated preparation timeline.
- 9 Use case ends successfully.

Alternate Flow – 5a. Table No Longer Available

- 5a1. During validation, the system finds the selected table was reserved by another diner in the same window moments earlier (overlapping booking).
- 5a2. The system displays a "Table just became unavailable" message and refreshes the floor plan to show current availability.
- 5a3. The diner is prompted to select a different Open table.
- 5a4. If the diner selects another table, the flow resumes at step 4 of the main scenario. If the diner cancels instead, the use case ends without creating a reservation.